SYNTHETIC CONFERENCE ORAL PRESENTATION · 10-MINUTE TALK

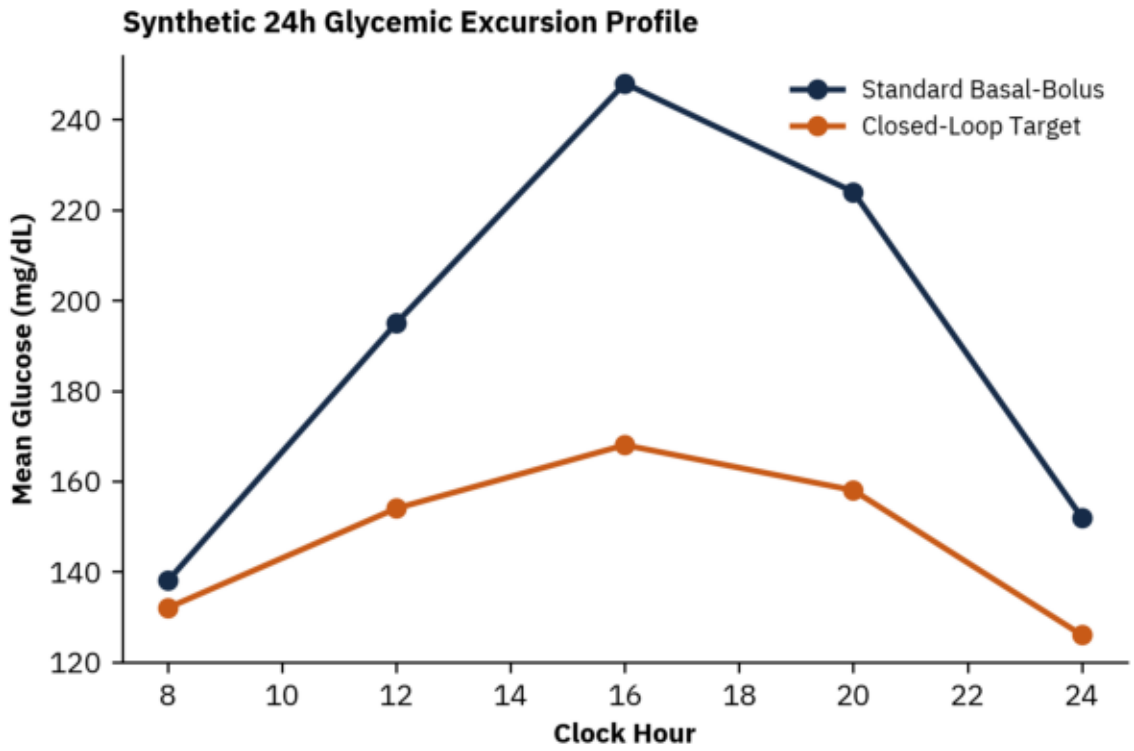
Closed-Loop Automated Insulin Delivery Doubles Time-in-Range During Inpatient Corticosteroid Therapy

A Synthetic Multicenter Randomized Controlled Trial (N = 280)

Synthetic Study Investigators · Department of Endocrinology & Biomedical Engineering

High-Dose Glucocorticoids Trigger Severe Afternoon Hyperglycemia Missed by Sliding-Scale Insulin

- Prednisolone >= 30 mg/day induces acute postprandial hepatic & peripheral insulin resistance
- Conventional basal-bolus regimens achieve < 45% glucose time-in-range (70–180 mg/dL)
- Reactive correction boluses increase nocturnal hypoglycemia (< 70 mg/dL)



280 Medical Ward Patients Were Randomized 1:1 to CGM-Guided Closed-Loop vs Basal-Bolus Care

1. Eligibility

Adults on >= 30 mg prednisolone-eq with 2 BG > 180 mg/dL across 6 hospitals.

2. 1:1 Randomize

Stratified by prior T2D diagnosis and steroid dose tier (n=140 per arm).

3. 10-Day Intervention

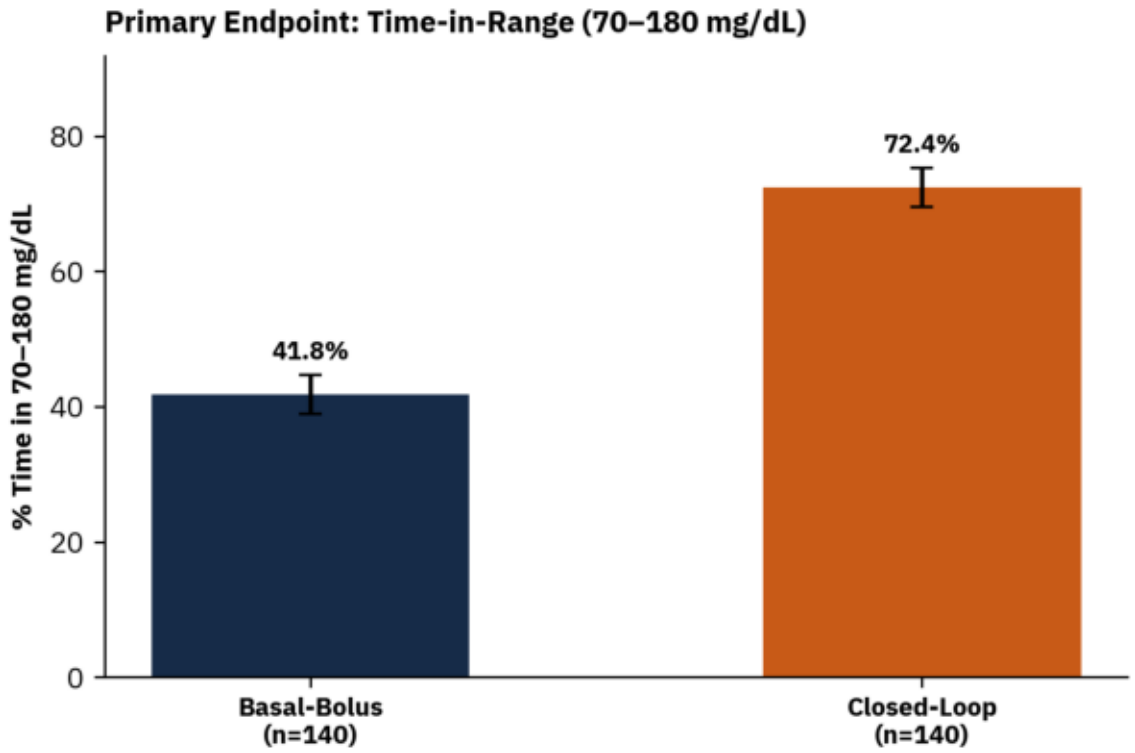
MPC closed-loop pump + CGM q5m vs masked-CGM standard basal-bolus.

4. Primary Endpoint

% Time-in-Range (70–180 mg/dL) over days 1–10 with safety hypoglycemia audit.

Closed-Loop Delivery Increased Time-in-Range From 41.8% to 72.4% Without Raising Hypoglycemia

- Mean difference +30.6 percentage points (95% CI +26.4 to +34.8; P < 0.0001)
- Mean sensor glucose fell from 204 mg/dL to 156 mg/dL
- Time < 70 mg/dL remained low in both arms (0.6% vs 1.4%; P = 0.03)



SYNTHETIC PRIMARY ENDPOINT · Error bars denote 95% CIs

Glycemic Control Improved Consistently Across Secondary Safety and Nursing Workload Metrics

Outcome Metric (Days 1–10)	Closed-Loop (n=140)	Basal-Bolus (n=140)	Difference / Effect	P-Value
Time > 250 mg/dL (%)	5.4 ± 4.1%	24.8 ± 9.6%	-19.4% (-21.1 to -17.7)	< 0.0001
Time < 70 mg/dL (%)	0.6% (IQR 0.0–1.1)	1.4% (IQR 0.3–2.6)	-0.8% (-1.2 to -0.3)	0.03
Severe Hypoglycemia (<54 mg/dL)	0 / 140 (0.0%)	2 / 140 (1.4%)	RD -1.4%	0.50
Fingerstick Checks / Patient-Day	1.2 ± 0.4	4.3 ± 0.8	-3.1 checks/day	< 0.0001

Conclusions & Honest Study Limitations

- Automated closed-loop insulin safely adapts to dynamic corticosteroid-induced insulin resistance on general wards

- Achieved +30.6% higher Time-in-Range while cutting capillary fingerstick burden by 72%

- LIMITATION: Open-label device wear; excluded ICU vasopressor patients and end-stage dialysis

- NEXT STEP: Cost-effectiveness & hospital length-of-stay powered pragmatic rollout